YONGJAE LEE

yj.lee@hpi.de | yongzzai.com

Google Scholar | GitHub | LinkedIn

Berlin, Germany

BIOGRAPHY

Yongjae Lee is a Ph.D. student in Computer Science at the Hasso Plattner Institute, University of Potsdam, Germany. He holds a M.Sc. in Industrial Data Science & Engineering and a B.Sc. in Industrial Engineering from Pusan National University, South Korea.

Research Interests

- Business Process Management - BPM
- Process-Aware Agentic Systems
- Deep Learning for BPM

EXPERIENCE

- **Data Systems, HPI [ ]** *2026.08 - Present*
  Scientific Assistant Potsdam, Germany
  - Conducting research on Agentic Business Process Management, and Data Systems
- **Industrial Artificial Intelligence Research Institute [ ]** *2026.03 - 2026.06*
  Research Associate Busan, South Korea
  - Collaborated with industry partners, including LG Electronics
  - Implemented deep learning models for industrial applications
- **Big Data Analytics Engineering Lab [ ]** *2022.09 - 2026.02*
  Researcher Busan, South Korea
  - Conducted research on Business Process Management, and Deep Learning
  - Led 3-year research projects as a main researcher (budget, reports, and development)

EDUCATION

- **Hasso Plattner Institute (University of Potsdam)** *2026.08 - Present*
  Ph.D. in Computer Science Potsdam, Germany
  - Supervisor: *Prof. Dr. Matthias Weidlich*
- **Pusan National University** *2024.03 - 2026.02*
  M.Sc. in Industrial Data Science & Engineering Busan, South Korea
  - Supervisor: *Prof. Dr. Hyerim Bae*
  - Thesis: *DHiM: A Dual-Modality Hierarchical Network for Multi-Perspective Business Process Anomaly Detection*
- **Pusan National University** *2018.03 - 2024.02*
  B.Sc. in Industrial Engineering Busan, South Korea

PROJECTS

- **Development of personalized smart operation logic based on customer product usage logs via process mining** *2025.03 - 2026.06*
  Funded by LG Electronics, Changwon
  - Developed a personalized product control system using Process Mining and Multimodal AI techniques, aiming to enhance user experience.
- **A Study on the XPL (eXplainable Process Learning) based on Artificial Intelligence** *2023.03 - 2026.02*
  Funded by the National Research Foundation of Korea (NRF)
  - Developed an explainable process learning methodology for process monitoring, detection, analysis, and improvement, based on artificial intelligence techniques.
- **Human Centered - Carbon Neutral Global Supply Chain Research Center** *2023.06 - 2025.02*
  Funded by the National Research Foundation of Korea (NRF)
  - Engaged in a national research project to develop key technologies for human safety and environmental sustainability across integrated global maritime supply chains.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT

- **[J.2]** Park, H., Sim, S., Park, K., **Lee, Y.**, and Park, E., and Bae, H. (2026. August) Hybrid Temporal Autoencoder and Similarity Matching for Low Aggregation Level Long Time-Series Forecasting. *Journal of Forecasting, 1–32.*
- **[P.2]** Park, T., **Lee, Y.**, Kim, D., and Bae, H. (2026, May). LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models. *arXiv preprint arXiv:2605.11011.*
- **[C.4]** **Lee, Y.**, Park, E., Park, D., Kim, D., Choi, J., and Bae, H. (2026, May). Process-Aware Procurement Lead Time Prediction for Shipyard Delay Mitigation. *In International Conference on Production Research (pp. 3-13). Cham: Springer Nature Switzerland.*
- **[C.3]** Park, T., **Lee, Y.** (co-first), Park, D., Kim, D., and Bae, H. (2025, December). JustDense: Just using Dense instead of Sequence Mixer for Time Series analysis. *In 2025 IEEE International Conference on Big Data (BigData) (pp. 1172-1184). IEEE.*
- **[P.1]** **Lee, Y.**, Park, T. (co-first), Sim, S., and Bae, H. (2025, October). Rewarding Structural Conformance of Reasoning using Process Mining. *arXiv preprint arXiv:2510.25065.*
- **[C.2]** **Lee, Y.**, Kim, D., Kim, D., and Bae, H. (2025, August). Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies. *In International Conference on Business Process Management (pp. 361-378). Cham: Springer Nature Switzerland.*
- **[J.1]** **Lee, Y.**, Park, K., Lee, H., Son, J., Kim, S., and Bae, H. (2024, December). Identifying key factors influencing import container dwell time using eXplainable Artificial Intelligence. *Maritime Transport Research, 7, 100116.*
- **[C.1]** Iman, A. N., Kamal, I. M., Ramadhan, M. H., Kim, D., **Lee, Y.**, and Bae, H. (2024, August). Predictive process monitoring for remaining time prediction with transfer learning. *ICIC Exp. Lett., 18(8), 851-858.*

CONFERENCE PRESENTATIONS

- **Korea Institute of Industrial Engineers 2025 Fall Conference** — *2025.11*
  DHiM: A Dual-Modality Hierarchical Network for Multi-Perspective Business Process Anomaly Detection — Gyeongju, South Korea
- **The 23rd International Conference on Business Process Management (BPM2025)** — *2025.09*
  Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies — Seville, Spain
- **The 28th International Conference on Production Research (ICPR28)** — *2025.07*
  Process-Aware Procurement Lead Time Prediction for Shipyard Delay Mitigation — Bogota, Colombia
- **The 11th International Conference on Logistics and Maritime Systems (LOGMS2023)** — *2023.09*
  Import Container Dwell Time: Analysis of Determinant Factors with Explainable Artificial Intelligence — Busan, South Korea

SKILLS

- **Programming Languages:** Python, JavaScript
- **Deep Learning Frameworks:** PyTorch, PyTorch Geometric, TRL
- **Markup & Styling:** HTML/CSS, LaTeX, Markdown
- **Database:** MySQL, Neo4j
- **Other Tools & Technologies:** Plant Simulation (Siemens)

HONORS AND AWARDS

- **Grand Prize** — *August 2023*
  PNU Industrial AI Competition, Pusan National University
- **Grand Prize** — *February 2023*
  Graduation Project, Dept. of Industrial Engineering, Pusan National University

LANGUAGES

Korean (Mother Tongue), English (Proficient), Chinese (Conversational)